

Four Fixes, One Uplink

A child tracker on 2G in the Philippines: four layered position sources, one bursty HTTPS uplink, and SMS as the channel that cannot be killed. *The design turns on a single deadline — transmit at five seconds, whatever the fix looks like.*

THE CONSTRAINT THAT SHAPES EVERYTHING

GNSS does not work indoors, and indoors is where children spend their days. Satellite signal sits near -130 dBm and buildings attenuate it 10–30 dB; through a school's concrete you get nothing at all. A-GNSS does not rescue this — assistance data cuts time-to-first-fix, it does not improve sensitivity. So GNSS is demoted to one source among four, Wi-Fi scanning becomes the indoor primary, and the firmware is built around a transmit deadline rather than a fix quality target.

REVISION NOTE

Why Find My Device was cut

REMOVED IN V3

v1 used Google's Find My Device network as the only channel; v2 kept it as a free fallback beneath a cellular primary. Both were wrong, and “add a free backup radio” is a tempting enough instinct to be worth recording why.

1. **Its coverage is a subset of what it was meant to back up.** Detection needs nearby Android phones opted into the network — often several, since the setting

commonly defaults to high-traffic areas only. Places dense with phone-carrying people are overwhelmingly places with cell coverage. It is weakest exactly where a fallback is needed.

2. **Its one real niche is already covered.** Dense indoors with poor cellular is genuine FMDN territory — and it is precisely what the Wi-Fi scan does better, with no third party and no extra radio.

3. **It introduces a problem that otherwise does not exist.** A device that advertises nothing trips no unwanted-tracker detection. Adding the beacon opts the child into “unknown tracker travelling with you” alerts on every nearby phone.

Also revised since: the SIM7080G was out of stock, so this builds on a **SIM800L** — 2G only, which is the right call in the Philippines where 2G is widely deployed and LTE-M effectively is not. That substitution cost us TLS on the modem and two days of battery, and bought us SMS as a channel that cannot be killed.

What we give up by dropping FMDN: a tracker that is off, dead, or out of coverage is simply gone. FMDN answers “where did the *device* go” — a lost-property question, not a child-safety one. Mitigated by last-known position with an honest age, a remote-triggered buzzer, and a short-range find mode the parent turns on only while searching.

FIGURE 1

End-to-end system

Two devices, one server, and two channels out of the tracker that fail in completely different ways. The HTTPS path carries everything rich; the SMS path carries everything that must arrive.

— HTTPS burst · 200 is the ack

— SMS · no GPRS, no TLS, no server

---- Conditional · on cache miss only

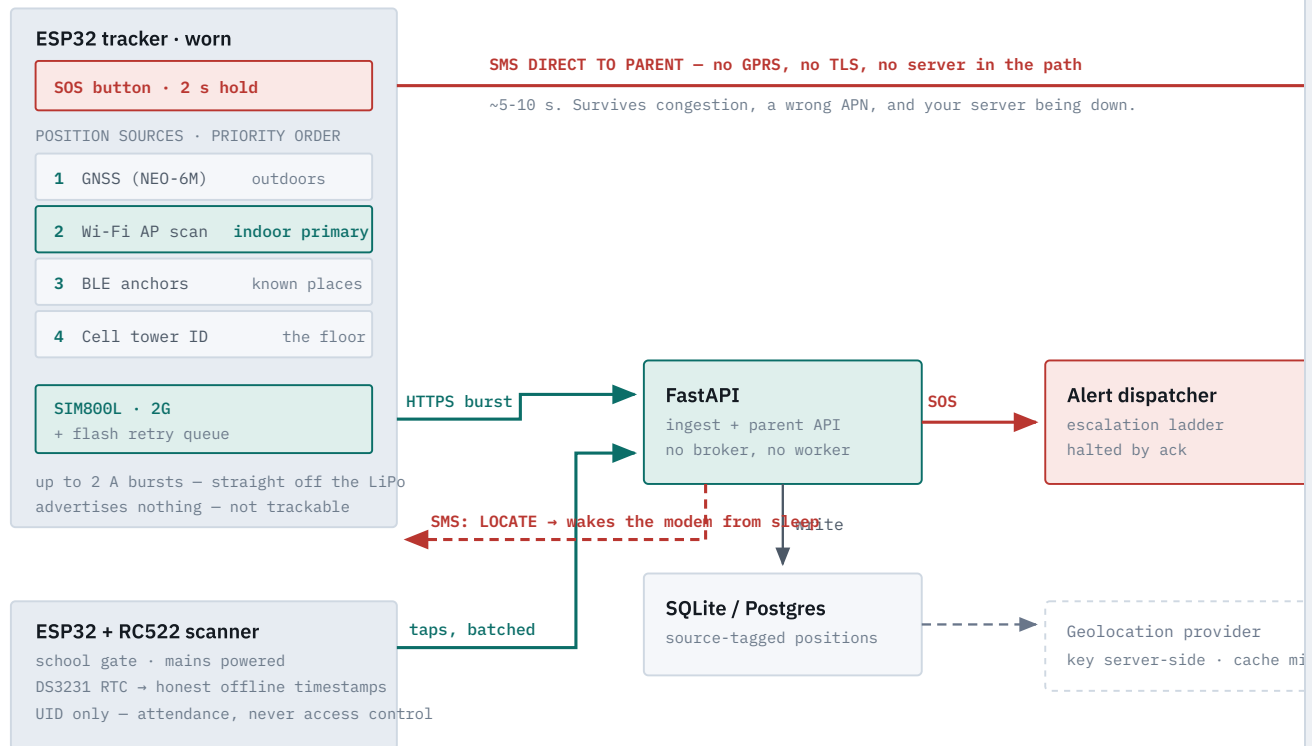


Figure 1. The two channels out of the tracker fail independently: HTTPS needs GPRS, a TLS handshake and a live server; the SMS needs none of them. That is why the SOS fires on both at once and the server skips its own text when the device reports it already sent one – one press must never arrive as two.

TRANSPORT

Why HTTP beat MQTT here

The plan called for MQTT. Then the GPRS session had to become bursty rather than persistent – carrier NAT drops an idle TCP session after 1–5 minutes, and the ~60 s keepalives needed to fight that stop the modem ever reaching `AT+CSCLK` sleep, the biggest power lever available. You would spend the battery keeping alive a socket the carrier kills anyway.

Once the connection is bursty, MQTT's persistent session buys nothing, and plain HTTPS is simpler in three concrete ways: **no broker** to run or secure, so the ingest worker collapses into a FastAPI route; **the 200 response is the ack**, removing the trap

where QoS 1 confirms delivery to the broker rather than to your server; and it is **debuggable with curl**, which on a 2G link you will want.

Two rules come with it. **Batch**, so the 3–8 s handshake is amortised over many points instead of paid per point. And give every event an **id**, so a retried POST is idempotent — a redelivery must never become a second position. One gotcha: TLS validation needs a roughly-correct clock and the ESP32 has no RTC, so pull the time from the modem with `AT+CCLK?` before the first handshake or every connection fails validation for no visible reason.

COVERAGE

What actually works, where

ENVIRONMENT	GNSS	WI-FI SCAN	BLE ANCHOR	CELL ID	RESULT
Outdoors, open sky	5–15 m	—	—	—	excellent
Urban street	10–30 m	15–40 m	—	—	good
Home / apartment	poor–none	10–30 m	exact place	—	good
School, office	none	15–40 m	exact place	—	good
Mall, transit hub	none	20–50 m	—	—	usable
Parking garage, basement	none	sparse	—	100 m–2 km	weak
Rural indoors	none	no APs in DB	exact place	1–5 km	poor

The uplink survives where the fix does not. LTE-M and especially NB-IoT were designed for basement utility meters — NB-IoT's coverage-enhancement modes buy roughly 20 dB of link budget over GSM. The SOS and the offline heartbeat keep working in places where a phone shows one bar. That asymmetry is the main argument for LTE-M over a plain LTE modem.

Known-place anchoring solves cost, latency and privacy at once — and the parent sets it up themselves, from the dashboard. See Figure 2.

Where it genuinely fails: underground parking, and rural interiors whose one access point is in no database. Both fall through to cell-tower ID — kilometres, not metres.

There is no clean fix, so the honest response is to return `source='cell'` with a truthful radius and draw a two-kilometre circle. Never let the map imply precision the fix does not have.

FIGURE 2

The parent names the place

The schema always had a slot for “these networks are school.” What it never had was a way to fill it — and asking a parent to type in MAC addresses was never going to happen. Showing them `DepEd-Classroom-3` and letting them tick it does.

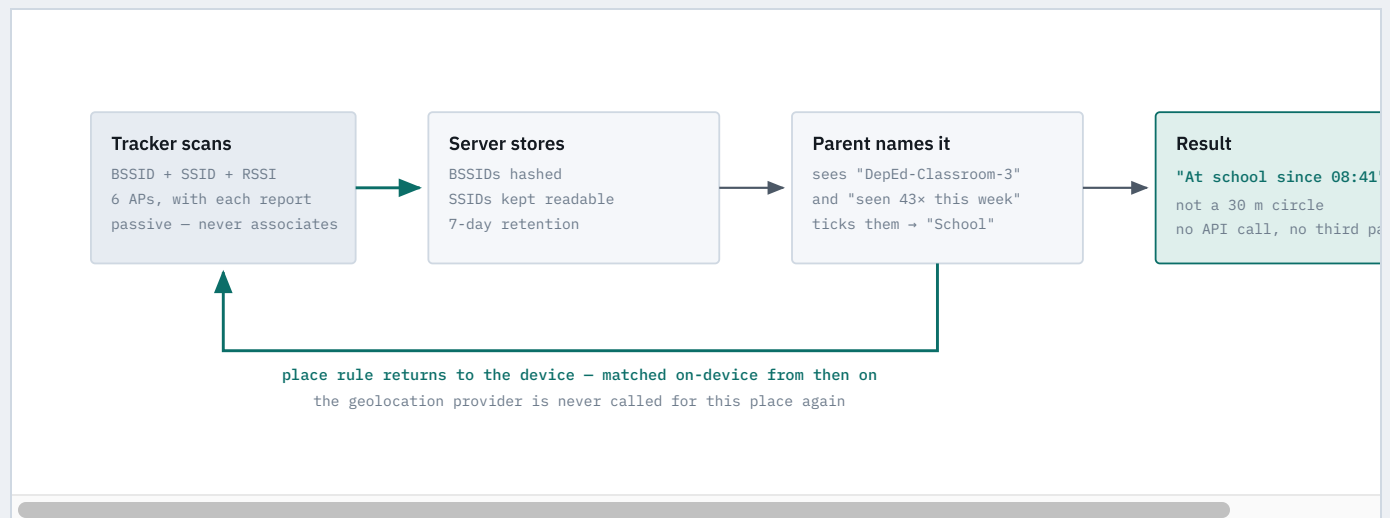


Figure 2. The loop closes on the device. Once a place is named, the matching happens in firmware against stored hashes — which is simultaneously the cheapest, the fastest and the most private outcome, because the provider never sees the networks the child sits inside every day.

BSSIDs are stored hashed; SSIDs are not. An access-point history is a finer-grained record of where a child has been than the coordinate history sitting next to it, so a database dump must not hand one over. But the parent needs something they can recognise to click on, and the matching runs on the hashes either way.

FIGURE 3

The five-second rule, indoors and out

The instinct is to wait for a good GNSS fix before transmitting. Outdoors that costs ten seconds. Indoors it costs everything, because the GNSS branch never returns at all.

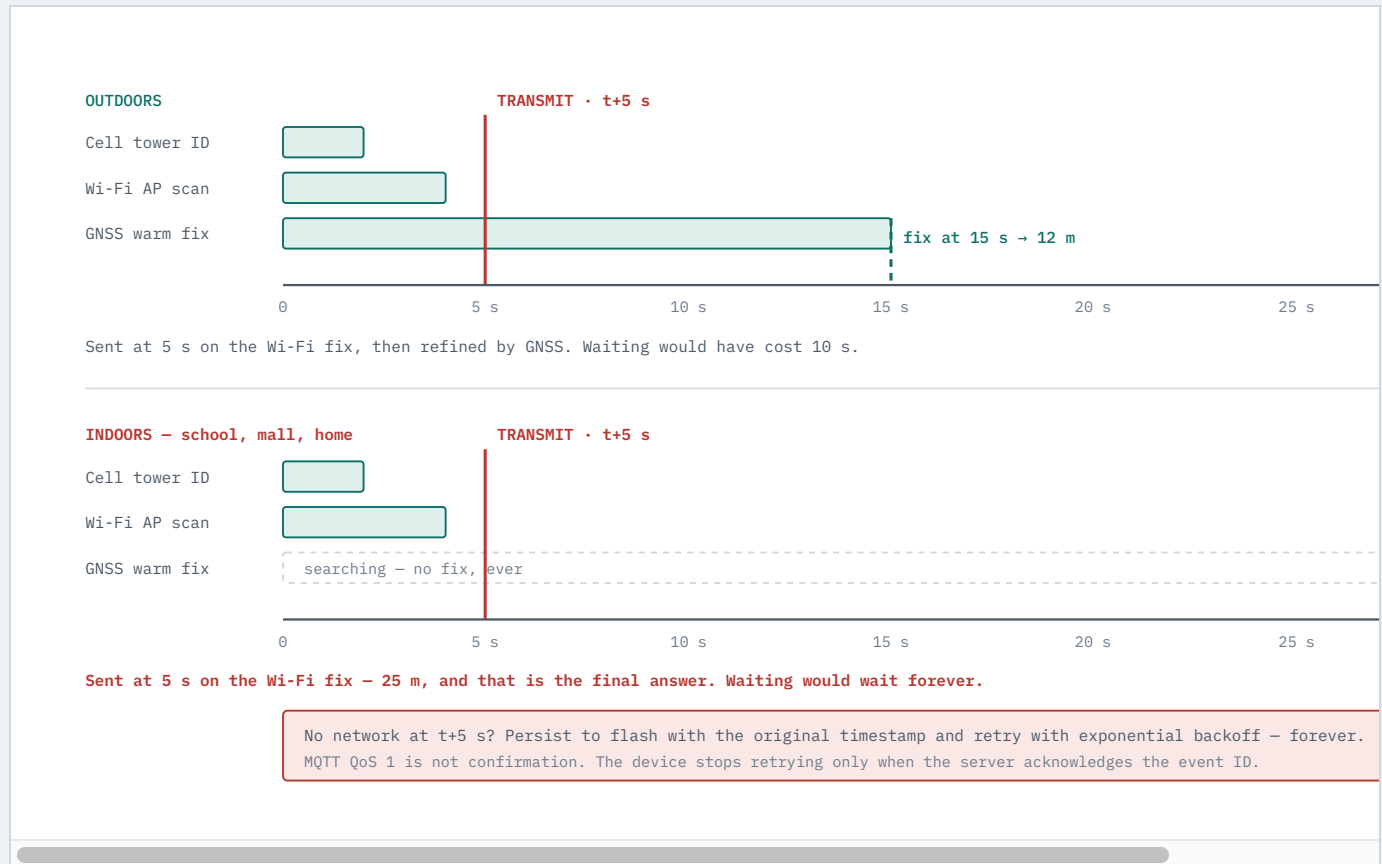
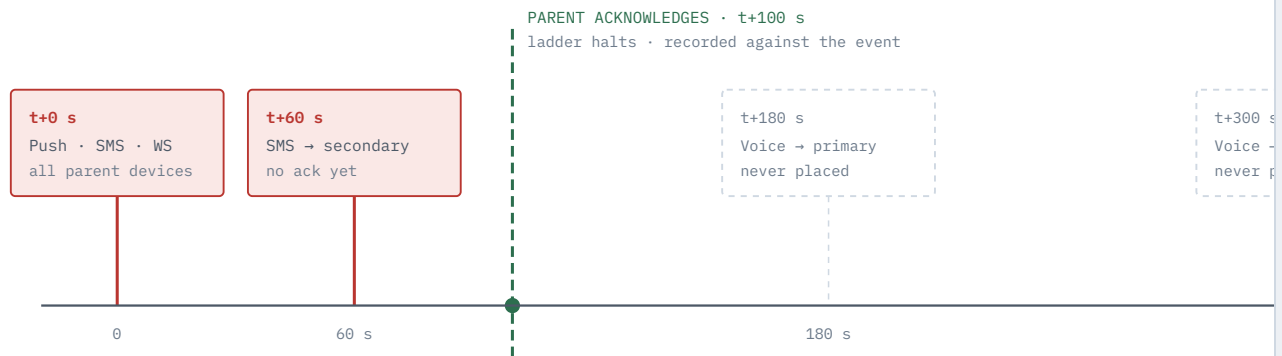


Figure 3. Same press, same deadline, different winning source. The deadline is not an optimisation – it is what makes the device work indoors at all. A design that waits for fix quality is a design that never fires in a school.

FIGURE 4

Escalation, not fan-out

Sending every channel at once and hoping is not a design. The ladder assumes the first parent may be asleep, driving, or out of signal, and keeps climbing until a human acknowledges – at which point it stops dead and records who answered.



Every message carries a one-tap deep link to the live map, not bare coordinates.
Rate limit: 5 SOS/hour/device – suppressed events still raise a "possible stuck button" operator alert.

Figure 4. The mechanism is the cut, not the ladder. Without an acknowledgement endpoint, escalation either runs to completion every time – training parents to ignore it – or is inferred from delivery receipts, which prove nothing about whether a human read it.

POWER MODEL

Report on state change, not on a timer

A fixed interval wastes the modem when the child has sat in a classroom for four hours, and is far too slow when they are in a car. Geofences are evaluated on-device, so a “left school” alert never waits for the next upload.

There is no accelerometer in the BOM, so motion is inferred from the radios already present: the serving cell every 60 s (free – the modem is awake anyway), and a Wi-Fi scan every 2–5 min compared against the last by Jaccard similarity. That scan is the authority, and it costs about 0.5% of the battery per day.

This is arguably the *better* signal. An accelerometer answers the wrong question – it reports that the device is being jostled, which a child fidgeting at a desk does all day, and every false “moving” burns a modem wake. A changed Wi-Fi neighbourhood reports that the child is somewhere *else*, which is what the cadence actually cares about. What is genuinely lost is interrupt-driven wake from sleep, which matters little when the modem outdraws the sleeping MCU by three orders of magnitude.

STATE	DETECTED BY	CADENCE
Stationary	Wi-Fi BSSID set unchanged (Jaccard ≥ 0.55), same serving cell	1 / 30 min
Moving	BSSID set changed, or serving cell changed	1 / 2 min
No usable signal	No APs visible <i>and</i> no cell change — rural	Fixed 1 / 10 min — never guess “stationary”
Geofence crossing	On-device polygon test	immediate
SOS active	Button	1 / 30 s for 30 min
Low battery < 15%	Fuel gauge	Alert, then 1 / hr

Target on a 2000 mAh cell is five days, charged every three or four. Measure with a PPK2 or INA219 — datasheet figures will not survive a real TLS handshake. Push a low-battery alert at 20% *and* 10%: a dead tracker still showing its last pin is the worst failure this system can produce.

BILL OF MATERIALS

Prototype hardware

COMPUTE ESP32 (WROOM-32) Supplies Wi-Fi and BLE scanning itself — two of the four position sources, at no extra cost.	UPLINK SIM800L 2G. Correct for the Philippines, where LTE-M and NB-IoT are not meaningfully available.
GNSS NEO-6M Keep V_BCKP powered, or every fix becomes a 27 s cold start instead of a ~1 s hot one.	POWER 2500–3000 mAh LiPo Bigger than planned: no LTE-M sleep mode means 2–3 days, not five.
NON-NEGOTIABLE	FEEDBACK

1000–2200 μF bulk cap SIM800L pulls 2 A in bursts. Without this, the rail browns out and the ESP32 resets — which reads as a firmware bug for days.	LED (<i>motor out of stock</i>) Same GPIO the motor will use. A buzz is felt through a pocket; an LED must be looked at — so the ack cue repeats for 20 s instead of firing once.
SCANNER ESP32 + RC522 Gate attendance station. 3.3 V only — 5 V destroys the reader. UID only, so attendance, never access control.	SCANNER CLOCK DS3231 RTC School Wi-Fi drops. A buffered tap with a wrong timestamp silently corrupts the record.

SEQUENCE

Build order, with gates

Phase 0 is the only real go/no-go in the project, and it now carries two gates: the uplink and the indoor fix. Both are a weekend with a bare module. Both are ruinous to discover in Phase 3.

00	Verify 2G coverage on <i>both</i> Globe and Smart, <i>and</i> Wi-Fi geolocation accuracy, on the child's real routes — indoors and out	GATE · <code>AT+CREG?</code> and <code>AT+CSQ</code> pass on the school run; Wi-Fi resolves to ≤ 50 m inside the school; SIM registered under RA 11934
01	SOS end-to-end: button $\rightarrow$ cellular $\rightarrow$ dispatcher $\rightarrow$ SMS, push and voice, with acknowledgement	GATE · p95 press-to-notification ≤ 20 s over 50 presses; 100% delivery from a dead zone; 20 indoor presses yield a usable fix; zero false SOS in 14 days of wear
02	Layered position fusion, known-place anchoring, BLE anchors, adaptive cadence, geofences	GATE · ≥ 5 days runtime; correct place detection at home and school for a week
03	FastAPI, schema, Leaflet dashboard, auth with 2FA, PWA push	GATE · No unauthenticated endpoint; per-device grants enforced in every query; WebSocket authenticated
04	RFID attendance scanner at the gate, with offline tap buffering	GATE · Taps survive a Wi-Fi outage with correct RTC timestamps; school network

05

Enclosure, field trial, SOS drills with the child

GATE · 30-day trial, no missed SOS; the child explains the confirmation patterns unprompted

FAILURE MODES

Silence is the enemy

Every failure below has the same shape: the system stops knowing where the child is, and the UI keeps showing the last pin as though nothing happened. Each needs an active alert, not a greyed-out dot.

FAILURE	DETECTED BY	RESPONSE
Battery dead	No telemetry for 2× expected interval	Push “tracker offline since 14:20”
No cell coverage	Modem detach	Queue in flash; show last known with its true age
No position source	All four sources fail or return cell-only	Report <code>source='cell'</code> with an honest radius — never fake precision
Geolocation provider down	API 4xx/5xx or quota	Fall back to anchors + cell; operator alert; positions still flow
Server down	External uptime monitor	Page the operator; device queues and replays
Prepaid load exhausted	GPRS attach fails; balance poll	Warn the parent <i>before</i> zero — in PH this is a routine, silent outage
SIM deregistered (RA 11934)	Registration refused	Operator alert — it looks exactly like a hardware fault
Stuck button	> 5 SOS/hour	Rate-limit plus a distinct operator alert

FAILURE	DETECTED BY	RESPONSE
Device left behind	No motion at a known place for hours	Optional “possibly not with the child” notice
Device lost or stolen	—	Revoke the device certificate; remote key wipe
Clock skew on device	Server-side timestamp check	Trust <code>received_at</code> ; flag the record

BEFORE PHASE 00

Two things that still bite

Conflating position sources is the most dangerous bug available here. `/api/devices` must return `source` , `accuracy_m` and `recorded_at` alongside the coordinates, and the map must never render a two-kilometre cell estimate the way it renders a four-second GNSS fix. Age badge on every pin, ring sized by real accuracy, desaturate past 15 minutes — and where a known place matches, show the place instead of the pin.

Tell the child. Dropping the beacon removed the anti-stalking problem but not the human one. A tracker a child knows about is a safety tool; one they discover is surveillance, and the discovery destroys trust irreparably. Age-appropriate disclosure, let them see their own map, and get the other parent's agreement in writing — in shared-custody situations, unilateral tracking has been treated as harassment.

Companion to PLAN.md · revision 4. A live database of a child's real-time position is among the highest-value targets a self-hosted server can carry — per-device mTLS, Argon2id with mandatory 2FA on parent accounts, a 90-day retention job that actually runs, the geolocation API key server-side only, and no port-forwarding from a home router.

If this becomes a deliverable for a client rather than your own child,
get real legal review before anyone else's child wears it.